

Using End Behavior and Multiplicity to Sketch Polynomials

Answer Key

Precalculus

Quick Answers

- | | |
|---|---|
| 1. $x(x-2)(x+2)$ | 6. $(x-1)(x+1)(x-2)(x+2)$ |
| 2. $-\infty$ | 7. (a) as x goes to minus infinity $= \infty$, — |
| 3. (a) zeros $= 1, -3$, — | 8. (a) factored form $= (x+2)(x-3)(x+3)$, |
| 4. (a) as x goes to minus infinity $= \infty$, (b) as x goes to plus infinity $= \infty$ | (b) zeros $= -2, 3, -3$ |
| 5. (a) y -intercept $= 2$, — | 9. (a) zeros $= 0, 2, -2$, — |
| | 10. (a) leading coefficient $= 1$, — |
-

What is verified

12 of 17 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 3, 5, 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSA-APR.B.3 – problems 1, 3, 6, 8, 10

HSF-IF.C – problems 2, 4, 5, 7, 9

Difficulty 1–5 of 5 across 17 tagged checks.

- 1.** Factor $p(x) = x^3 - 4x$ completely.

Solution: Take out the common factor x , then recognise a difference of squares: $x^3 - 4x = x(x^2 - 4)$.

$$p(x) = x(x-2)(x+2)$$

- 2.** For $p(x) = -2x^3 + 5x - 1$, state $\lim_{x \rightarrow \infty} p(x)$.

Solution: Only the leading term matters as x grows: $-2x^3$ with x large and positive is large and negative. The lower-order terms cannot catch up.

$$\lim_{x \rightarrow \infty} p(x) = -\infty$$

- 3.** $p(x) = (x-1)^2(x+3)$. (a) Zeros. (b) Cross or touch.

Solution (a): Setting each factor to zero gives $x = 1$ (from the squared factor) and $x = -3$.

$$x = 1 \quad \text{and} \quad x = -3$$

Solution (b): *Instructor-judged.* The graph crosses at -3 and only touches at 1 . The reason must come from the exponents: the factor $(x + 3)$ appears to the first power — odd multiplicity — so the product changes sign there; $(x - 1)^2$ has even multiplicity, so the sign does not change and the curve turns back on itself. Naming the two zeros correctly without the parity reason is half the answer.

4. $p(x) = 3x^4 - x^2 + 2$. (a) Limit at $-\infty$. (b) Limit at $+\infty$.

Solution (a): The degree is even and the leading coefficient is positive, so both ends rise. As $x \rightarrow -\infty$, $3x^4 \rightarrow +\infty$.

$$\lim_{x \rightarrow -\infty} p(x) = \infty$$

Solution (b): Same leading term, same conclusion at the other end.

$$\lim_{x \rightarrow \infty} p(x) = \infty$$

5. $p(x) = (x + 2)(x - 1)^2$. (a) y -intercept. (b) Sketch.

Solution (a): $p(0) = (0 + 2)(0 - 1)^2 = 2 \cdot 1$.

$$p(0) = 2$$

Solution (b): *Instructor-judged.* A correct sketch crosses the x -axis at -2 (odd multiplicity), touches it at 1 without crossing (even multiplicity), passes through $(0, 2)$, falls to the lower left and rises to the upper right — the degree is 3, odd, with a positive leading coefficient. Mark down a sketch that crosses at 1 : the squared factor is the feature this item tests. Mark down a sketch whose two ends point the same way, which would require an even degree.

6. Factor $p(x) = x^4 - 5x^2 + 4$ completely.

Solution: Treat it as a quadratic in x^2 : $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$, and each bracket is a difference of squares.

$$p(x) = (x - 1)(x + 1)(x - 2)(x + 2)$$

7. $p(x) = -x^5 + 3x^2$. (a) Limit at $-\infty$. (b) Describe the end behavior.

Solution (a): As $x \rightarrow -\infty$, $x^5 \rightarrow -\infty$, so $-x^5 \rightarrow +\infty$ and the $3x^2$ term is negligible.

$$\lim_{x \rightarrow -\infty} p(x) = \infty$$

Solution (b): *Instructor-judged.* The curve rises without bound on the left and falls without bound on the right. The two deciding features are the odd degree, which makes the ends go opposite ways, and the negative leading coefficient, which makes the right end the falling one. An answer that names only one of the two has half the rule; citing $3x^2$ as relevant to end behavior is wrong.

8. $p(x) = x^3 + 2x^2 - 9x - 18$. (a) Factor. (b) Zeros.

Solution (a): Group in pairs:

$$\begin{aligned} x^3 + 2x^2 - 9x - 18 &= x^2(x + 2) - 9(x + 2) \\ &= (x^2 - 9)(x + 2) \end{aligned}$$

and $x^2 - 9$ is a difference of squares.

$$p(x) = (x + 2)(x - 3)(x + 3)$$

Solution (b): Each factor is linear, so every zero has multiplicity 1 and the graph crosses at all three.

$$\boxed{x = -3, \quad x = -2, \quad x = 3}$$

9. $p(x) = x^4 - 4x^2$. **(a)** Factor and list zeros. **(b)** Sketch.

Solution (a): $x^4 - 4x^2 = x^2(x^2 - 4) = x^2(x - 2)(x + 2)$, so the zeros are 0 (multiplicity 2), 2 and -2 .

$$\boxed{x = -2, \quad x = 0, \quad x = 2}$$

Solution (b): *Instructor-judged.* The sketch must cross at -2 and at 2 , touch at 0 without crossing, dip below the axis on both $(-2, 0)$ and $(0, 2)$, and rise at both outer ends — degree 4, even, positive leading coefficient. A sketch that crosses at 0 , or whose ends go opposite ways, misses the point of the item.

10. $p(x) = a(x + 1)^2(x - 3)$ with $p(0) = -3$. **(a)** Find a . **(b)** Sketch and state end behavior.

Solution (a): Substitute $x = 0$:

$$\begin{aligned} a(0 + 1)^2(0 - 3) &= -3 \\ -3a &= -3 \end{aligned}$$

So

$$\boxed{a = 1}$$

Solution (b): *Instructor-judged.* With $a = 1$ the rule is $p(x) = (x + 1)^2(x - 3)$. The sketch must touch the axis at -1 without crossing, cross at 3 , pass through $(0, -3)$, fall to the lower left and rise to the upper right. The stated end behavior must match that, and must be justified by an odd degree with a positive leading coefficient. Reject a sketch that crosses at -1 .